

Check fitted flocker model

Specify model to check:

```
spec <- params$spec #final_species_eco_sorted[6] # 1
spec
```

```
[1] "Ferruginous Hawk"
```

```
buffer_km <- params$buffer_km
buffer_km
```

```
[1] 750
```

```
fold <- params$fold
# fold

results_dir <- params$results_dir
results_dir
```

```
[1] "M:/Documents/DEBTs/analysis/Schifferle_BBS_occupancy_models_2023/results/full_model"
```

Load fitted model:

```
load(file = file.path(results_dir,
                       paste0("out_", spec, "_fm_buffer", buffer_km, ".RData"))) # xx adjust
#print(out)

posterior <- brms::as_draws(out)
```

Check whether MCMC worked:

Do we have divergent transitions?

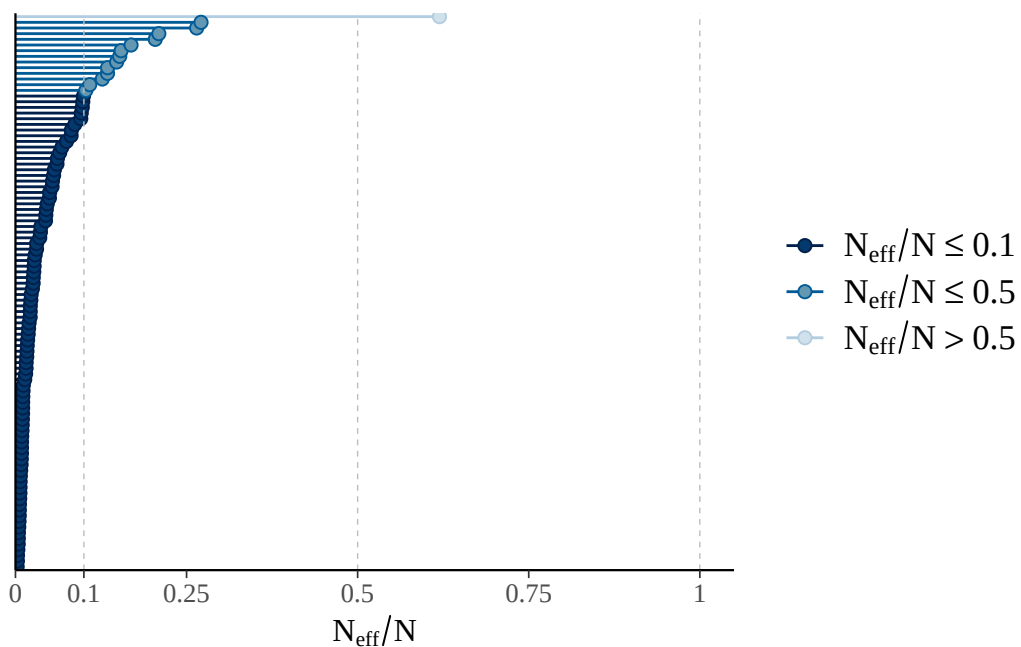
```
[1] "divergent transitions: 0"
```

Explorations regarding divergent transitions:

...

Is effective number of MCMC samples large enough:

The larger the ratio of n_{eff} to N (number of samples), the better.



```
[1] "effective sample size too small"
```

Warning: Parts of the model have not converged (some Rhats are > 1.05). Be careful when analysing the results! We recommend running more iterations and/or setting stronger priors.

```
[1] "n_eff ratio bulk ESS:"
```

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.002471	0.009222	0.022678	0.047338	0.060815	0.271125

[1] "n_eff ratio tail ESS:"

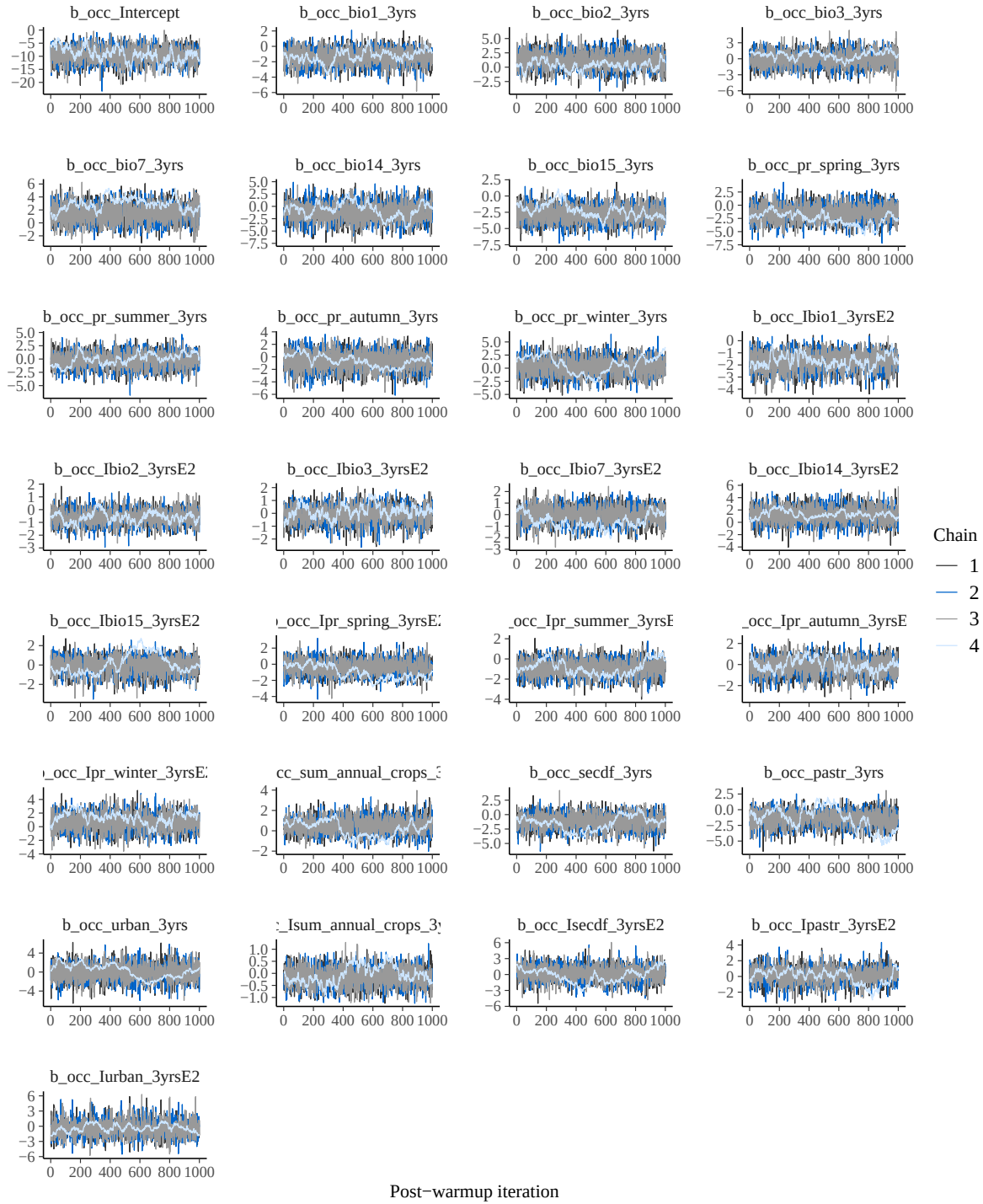
Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.005156	0.062275	0.183814	0.235550	0.390219	0.659958

[1] "effective sample size in bulk or tail too small"

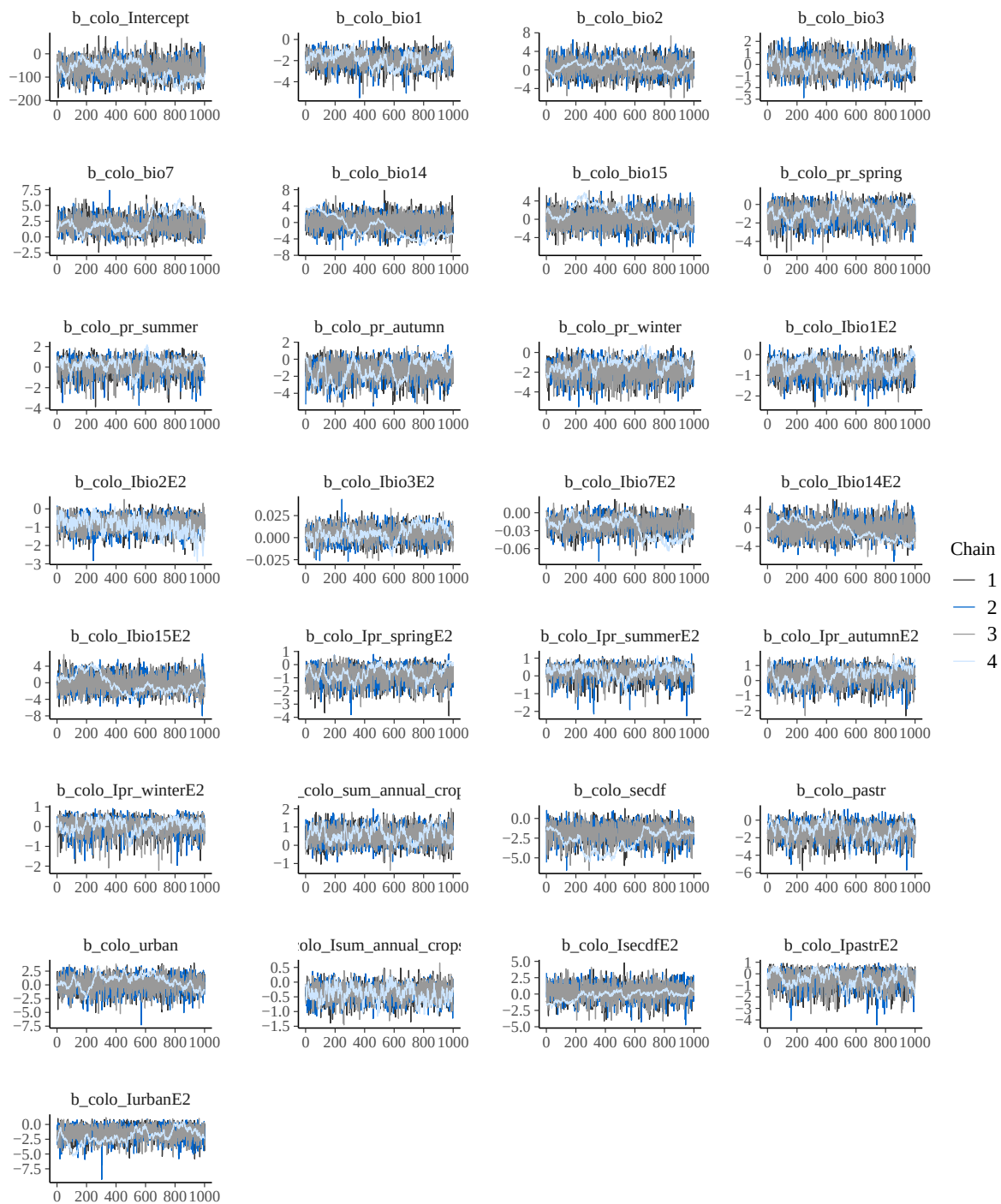
Do chains mix well:

No divergences to plot.

No divergences to plot.



No divergences to plot.



Post-warmup iteration

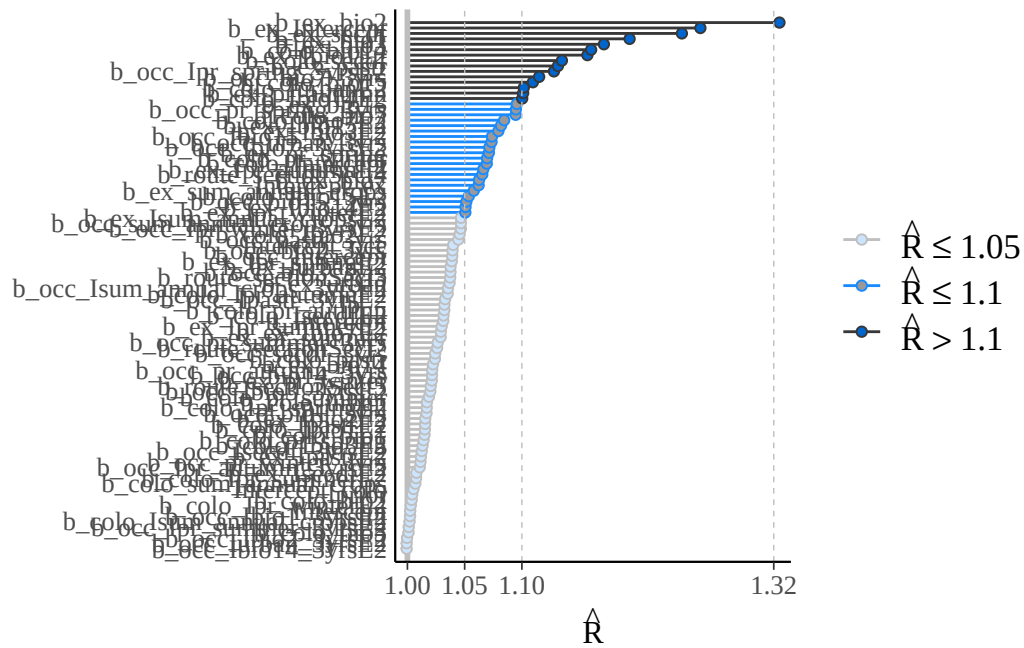


Post-warmup iteration

Are R-hat values fine:

Min.	1st Qu.	Median	Mean	3rd Qu.	Max.
0.9993	1.0159	1.0361	1.0535	1.0708	1.3250

```
[1] "R-hat > 1.05 for 36 parameters."
```



Print model posteriors

```
[1] "parameters for which uncertainty interval doesn't overlap zero:"
```

```
[1] "b_occ_Intercept"      "b_occ_bio15_3yrs"    "b_occ_Ibio1_3yrsE2"
```

```
[1] "b_colo_bio1"          "b_colo_pr_winter"    "b_colo_Ibio2E2"
```

```
[1] "b_ex_Ibio2E2"
```

